MIL-I-52172C
12 August 1983
SUPERSEDING
MIL-I-52172B
29 February 1968

MILITARY SPECIFICATION

INSULATION BATT, NYLON, THERMAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of nylon batt used for insulation purposes, which is inherently flame and fungus resistant.

* 1.2 Military part number. The military part number shall consist of the letter B, the basic number of this specification and a dash number specifying the number of total yards required.

Example:	B52172	-	70
	Basic number of MIL-SPEC		Number of total yards (see 3.4.2)

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:09Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

	V-T-295	Thread, Nylon
*	SS-S-118	Sound Controlling (Acoustical) Tiles and Panels
*	PPP-B-585	Box, Wood, Wirebound
*	PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated
*	PPP-B-601	Boxes, Wood, Cleated Plywood
*	PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

	MIL-P-116	Preservation; Methods of
*	MIL-F-22191	Barrier Material, Transparent, Flexible, Heat Sealable

STANDARDS

FEDERAL

*	FED-STD-191	Textile Test Methods
	FED-STD-751	Stitches, Seams and Stitchings

MILITARY

	MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
	MIL-STD-129	Marking for Shipment and Storage
	MIL-STD-130	Identification Marking for US Military Property
*	MIL-STD-810	Environmental Test Methods
	MIL-STD-831	Test Reports, Preparation of
*	MIL-STD-1188	Commercial Packaging of Supplies and Equipment

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:09Z
Check the source to verify that this is the current version before use.

* (Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM C177 - Thermal Conductivity of Materials by Means of the Guarded Hot Plate.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

* (Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article sample. When specified (see 6.2) before production is commenced, a sample of the material covered by this specification shall be submitted, or made available, to the procuring agency for approval. Approval of the first article sample authorizes the commencement of production, but does not relieve the supplier of the responsibility for compliance with all of the applicable provisions of this specification. The first article sample shall be manufactured in the same facilities to be used for manufacture of the production of items.

3.2 Materials.

3.2.1 Batt.

3.2.1.1 Composition. The insulation batt shall be 100 percent new, undyed nylon. A minimum of 95 percent of nylon shall be fully drawn and the filament length shall be sufficiently long to give the batt good cohesive properties. The size of the filament shall be under 10 denier.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:09Z
Check the source to verify that this is the current version before use.

3.2.1.2 Acidity. The finished batt shall have a pH of water extract of 6.0 to 8.0 when tested as specified in 4.6.

3.2.1.3 Weight. Unless otherwise specified by the procuring activity, the weight of the batt shall be 25 ounces ± 10 percent per square yard.

3.2.2 Mesh. The mesh for quilting the batt shall be white, 100 percent nylon. The size shall be 40 denier and the weight shall be 1/2 ounce ± 10 percent per square yard.

3.2.3 Thread. The thread shall be white nylon and shall conform to V-T-295 type I, class 2, size B.

3.3 Construction.

3.3.1 Stitching. The insulation batt shall be faced on both sides with nylon mesh and stitched with nylon thread, in a diamond or square pattern approximately 2 inches between sides of quadrilateral. Stitching shall conform to type 301 of Fed. Std. No. 751, 5 stitches to the inch. Thread tension shall be maintained properly to avoid loose stitching and the lock shall be embedded in the center of the batt. Thread breaks shall be back stitched not less than 1 inch at each break. Thread ends shall be trimmed.

3.4 Dimensions.

3.4.1 Width. The minimum width of the finished batt shall be 42 inches.

3.4.2 Length. Unless otherwise specified, the total length of roll of insulation batt shall contain not less than 35 linear yards nor more than 36 linear yards, (see 6.2). There shall be no more than 2 pieces to the roll and no piece shall be less than 6 yards in length.

3.5 Flame resistance. No flame shall issue from the specimen insulation batt during or after 10 seconds of flame exposure when subjected to the flame resistance test of 4.6. Glow shall not progress beyond the area covered by the test flame.

3.6 Fungus resistance. The insulation batt shall not exhibit any fungus growth when tested in accordance with 4.6.

3.7 Thermal conductivity (K factor). The approximate thermal conductivity of the insulation batt shall not exceed $\frac{0.30 \text{ BTU inch}}{\text{h. X sq. ft. X } ^{o}\text{F}}$ when tested as specified in 4.6.

3.8 Identification. Each roll or bolt of the insulation batt shall be marked for identification in accordance with MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

3.9 Workmanship. The finished insulation batt shall contain no stains or streaks, holes, cuts or tears or weak places in the mesh (see 4.5.3).

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all insulation batt produced under essentially like conditions and offered for acceptance at one time. The size of lot shall be expressed in linear yards.

4.3 Classification of inspection. The inspection of the insulation batt shall be classified as follows:

(a) First article inspection.
(b) Acceptance inspection.

4.3.1 First article inspection. The first article inspection shall consist of all the examination and tests specified herein. When specified, a first article sample of 1 yard full width of material shall be accompanied by a test report and submitted to the procuring activity (see 6.2) for approval. The test report shall cover all tests of 4.6. For Air Force procurement the test report shall be prepared in accordance with MIL-STD-831.

4.3.2 Acceptance inspection. Acceptance inspection for acceptance of individual lots shall consist of all requirements of this specification.

4.4 Sampling.

4.4.1 For overall examination. Selection of rolls of insulation batt shall be made in accordance with the terms of MIL-STD-105. The number of rolls selected for examination shall be in accordance with table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:05Z
Check the source to verify that this is the current version before use.

Table I. Number of rolls in sample

Lot size in yards	Sample size in rolls
Up to 1300	3
1301 to 3200	5
3201 to 8000	7
8001 to 22,000	10
22,001 and over	15

4.4.2 For yard-to-yard examination. Rolls of insulation batt selected in accordance with 4.4.1 shall be used for yard-to-yard examination. The unit-of-product shall be one yard and the sample size shall be in accordance with level II of MIL-STD-105. An approximately equal number of yards shall be examined from each roll.

4.4.3 For tests. Samples of insulation batt shall be selected at random for tests. These units-of-product shall be full width of the batt and approximately 1 yard in length. The number of units-of-product shall be in accordance with table II.

Table II. Number of units-of-product for tests

Lot size in yards	Number of units-of-product
Less than 800 yards	3
801 to 22,000 yards	5
22,001 yards and over	7

4.5 Examination.

4.5.1 Overall examination. Rolls of insulation batt selected in accordance with 4.4.1 shall be examined for overall defects. Each defect listed in table III shall be counted no more than once for each roll examined. The unit-of-product for this examination shall be one roll. The number of acceptable defects shall be 0 for 10 rolls or less, and 1 for over 10 rolls.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

Table III. Overall defects

Category	Defect	Requirement paragraph	Method of inspection
Critical	None defined		
Major:			
101	Nylon not new, and undyed.....	3.2.1.1	Visual
102	Mesh not white nylon.....	3.2.2	Visual
103	Batt, not faced on both sides.....	3.3.1	Visual
104	Stitching not in proper pattern....	3.3.1	Measure
105	Stitches not 301 type.....	3.3.1	Visual
106	Width less than specified.....	3.4.1	Measure
107	Improper identification.....	3.8	Visual
108	Overall uncleanliness.....	3.9	Visual

4.5.2 Length.

4.5.2.1 Length of roll. Rolls of insulation batt selected in accordance with 4.4.1 shall be examined for the defects listed below. The maximum number of acceptable defects shall be 0 for 10 or less rolls, and 1 for more than 10 rolls.

(a) Length of roll less than 35 yards or over 36 yards.
(b) More than 2 pieces in roll.
(c) Roll containing piece less than 6 yards in length.

4.5.2.2 Total length of rolls. The lengths on the rolls for 4.5.2.1 shall be totaled and compared with the total length marked on the identification tags or labels. The lot shall be considered unacceptable if the measured total length is less than the total of the length marked on the tags or labels.

4.5.3 For yard-to-yard defects. Insulation batt selected in accordance with 4.4.2 shall be examined for the defects listed in table IV. Defects shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the batt, in which case only the more serious defects shall be counted. The acceptable quality level (AQL) for this examination shall be 3.0 major defects and 10.0 total defects (major and minor) per 100 units (yards).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

Table IV. Yard-to-yard defects

Category	Defects	Requirement paragraph	Method of inspection
Critical	None defined		
Major:			
109	Stains or streaks.....	3.9	Visual
110	Holes.....	3.9	Visual
111	Cuts or tears.....	3.9	Visual
112	Weak place in mesh.....	3.9	Visual
Minor:			
201	Not 5 stitches to the inch.....	3.3.1	Measure
202	Loose stitches.....	3.3.1	Visual
203	Lock in stitches not imbedded properly...	3.3.1	Visual
204	Thread backs not back stitched 1 inch....	3.3.1	Measure
205	Thread ends not trimmed.....	3.3.1	Visual

4.5.4 Packaging, packing and marking for shipment. Examination shall be made for compliance with the requirements of section 5.

* 4.6 Tests. Insulation batt selected in accordance with 4.4.3 unless otherwise indicated shall be tested in accordance with the method of FED-STD-191 as indicated in table V.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T21:09Z
Check the source to verify that this is the currentversion before use.

General Manager

Table V. Test methods

Characteristic	Requirement paragraph	FED-STD-191 Test method	Determinations per unit-of-product	Results reported as:
Weight Nylon batt	3.2.1.3	5041	5	Average of 5 determinations to nearest 0.1 ounce.
Nylon mesh	3.2.2	5041	5	Average of 5 determinations to nearest 0.1 ounce.
Denier	3.2.1.1	4021	1	Pass or fail
Batting filament	3.2.2	4021	1	Pass or fail
Mesh	3.2.1.1	1530	1	Pass or fail
Batt composition	3.2.1.2	2811	2	Average of 2 determinations to nearest 0.1 pH.
Acidity of batt (pH)				
Thread	3.2.3	1/	4	Average of 4 determinations to nearest whole number.
Flame resistance	3.5	2/	5	Pass or fail
Fungus resistance	3.6	3/	1	Growth or no growth
Thermal conductivity 3/	3.7	4/	2	Average of 2 determinations to nearest 0.01 Btu.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

1/ The thread shall be tested in accordance with V-T-295.

* 2/ The flame resistance test for the insulation batt shall be in accordance with flame resistance test method and equipment of SS-S-118 for Class A material except test shall be for 10 seconds.

* 3/ Tests for fungus resistance shall be in accordance with method 508.2 of MIL-STD-810 except that the duration of the test shall be 14 instead of 28 days.

4/ The thermal conductivity test for the insulation batt shall be in accordance with ASTM C177.

* 5. PACKAGING.

* 5.1 Preservation, packing, unitization and marking - Level A, B or Industrial (see 6.2).

* 5.1.1 Level A or B. Each roll shall be packaged Method IC-3 in accordance with MIL-P-116 by heat sealing into a transparent barrier bag made of type II, MIL-F-22191 plastic film.

* 5.1.2 Industrial. Preservation, packing, unitization and marking shall be in accordance with MIL-STD-1188.

* 5.2 Packing.

* 5.2.1 Level A or B. Uniform quantities of insulation roll batts packaged in accordance with paragraph 5.1 shall be overpacked in boxes conforming to PPP-B-621, class 1, style as applicable; PPP-B-601, domestic type; PPP-B-585, class 1 or 2; or PPP-B-591, domestic type.

* 5.2.2 Industrial. Packing shall be in accordance with MIL-STD-1188.

* 5.3 Marking. Marking shall be in accordance with MIL-STD-129 unless otherwise specified.

6. NOTES.

6.1 Intended use. The nylon batts covered by this specification are used in electrical and non-electrical insulation blankets for rockets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurment documents should specify the following:

(a) Title, number and date of this specification.
* (b) Military part number (see 1.2) including number of yards required (see 4.2).
* (c) Whether first article sample is required and if so place of delivery.
(d) Weight of batt if other than that specified (see 3.2.1.3).
(e) Place of inspection and tests required.
(f) Length of roll if other than that specified (see 3.4.2).
(g) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(h) Packing if other than specified (see 5.2.1).
(i) Special marking if required (see 5.3).

* 6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army-MI
Air Force-99

Review Activities:
Army-ME
Air Force-84

Preparing Activity:
Army - MI

Project No. 5640-0363

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:09Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Missile Command
ATTN: DRSMI-RSDS
Redstone Arsenal, AL 35898

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:09Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:09Z
Check the source to verify that this is the current version before use.